

NORTHERN UNIVERSITY OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering

CSE 4204 — Mobile Computing Lab | Instructor: Md. Riaz Mahmud, Assistant Professor

WEEK 02: PROJECT PROPOSAL

Official Team Name	CSE4204-8A-T07
Section	8A
Project Title	AI-Smart Campus System
GitHub Repository	https://github.com/nazmusshakib878/CSE4204-8A-T07-ai-smart-campus-system.git

PART A: Project Proposal

1. Project Title: AI-Smart Campus System

2. Problem Statement

Current Problem: Modern university students often face significant difficulties in managing fragmented academic information, tracking personal performance indicators such as CGPA and attendance, and staying up-to-date with institutional notices, routines, and academic deadlines. Faculty members also struggle to efficiently monitor student engagement, performance trends, and identify academically vulnerable students at an early stage. Most traditional Learning Management Systems (LMS) function primarily as passive information repositories. They provide limited automation, lack intelligent academic guidance, and fail to deliver personalized recommendations based on student performance data.

Importance of the Problem: An efficient academic management ecosystem is essential for improving student success, institutional productivity, and educational quality. This problem is important because:

- **Academic Decision Support is Limited:** Students often select courses without personalized academic guidance based on their strengths and previous results.
- **Delayed Performance Intervention:** Faculty members may identify struggling students too late, reducing opportunities for early academic support.
- **Fragmented Information Access:** Notices, routines, attendance records, and academic updates are often distributed across multiple sources.

- **Reduced Student Productivity:** Lack of centralized academic tools increases confusion, delays communication, and negatively impacts learning efficiency.
- **Growing Need for Intelligent Automation:** Modern educational institutions require smart digital systems capable of providing recommendations, predictions, and automated assistance.

Insufficient Existing Solutions : Existing campus portals and traditional Learning Management Systems (LMS) provide basic administrative and academic functionalities but offer limited intelligent capabilities. Most current systems fail to provide:

- AI-based personalized course recommendation.
- Automated student risk prediction and performance analytics.
- Intelligent campus-specific virtual assistance.
- Real-time academic decision support.

Although general AI tools such as Gemini or ChatGPT are available, they are not directly connected with institutional academic data, university routines, student performance records, or official notices. Therefore, they cannot provide accurate campus-specific academic guidance.

Relevant AI for solving the Problem: Artificial Intelligence plays a crucial role in transforming a conventional campus platform into an intelligent academic assistance ecosystem. The proposed

system integrates AI in the following practical ways:

1. AI-Powered Academic Course Advisor

A personalized recommendation system will analyze a student's previous academic records, strengths, and semester performance to recommend suitable elective and core courses for future semesters.

How it helps:

- Supports better academic planning.
- Reduces poor course selection decisions.
- Provides data-driven academic recommendations.

2. Student At-Risk Detection & Performance Analytics

An intelligent analytics mechanism will evaluate student engagement indicators such as:

- Attendance percentage

- Quiz marks
- Assignment performance
- Mid-term results

Based on predefined predictive conditions, the system will automatically generate High-Risk Alerts for students who are academically vulnerable. How it helps:

- Enables early intervention.
- Improves academic monitoring.
- Supports faculty decision-making.

3. AI Campus Assistant (Gemini API Integration)

The system will integrate an NLP-driven chatbot powered by Gemini API to provide instant responses regarding:

- University notices
- Exam schedules
- Library hours
- Class routines
- Campus FAQs

How it helps:

- Provides 24/7 automated assistance.
- Reduces dependency on manual inquiries.
- Improves accessibility to institutional information.

3. Objectives

- To build a smart, all-in-one student productivity portal connected to campus workflows.
- To generate personalized study paths and optimization strategies based on historical grading and performance trends.
- To integrate a 24/7 AI Academic Assistant (Chatbot) to solve institutional information access issues.
- To maximize student success and minimize academic dropout rates through predictive grade warnings.

4. Proposed Features

Feature	Description
Secure Authentication	Role-based authentication using JWT tokens for students and faculty.
Smart Performance Dashboard	Interactive visual charts tracking student attendance patterns, quiz distributions, and mid-term grades.
AI Academic Assistant	An embedded LLM interface trained on university rules, curriculum guidelines, and individual performance logs to deliver immediate study plans and answers.
Administrative Control Panel	User approval and management with platform control operations.
Predictive Risk Alerts	An automated warning notification system alerting students when attendance drops or poor KPIs indicates an overall grading risk.

5. Technology Stack

- ▶ **Frontend:** React.js, Tailwind CSS (Formulated for fully responsive mobile and web layouts)
- ▶ **Backend:** Laravel
- ▶ **Database:** MySQL (XAMPP)
- ▶ **AI Integration:** Gemini API / OpenAI API/ HuggingFace (Utilizing custom system profiling and prompt engineering)

6. Expected Outcome

User Gains

Students will gain:

- Personalized academic recommendations.
- Better academic planning support.
- Centralized access to academic information.
- Instant AI-powered campus assistance.

Faculty members will gain:

- Efficient student monitoring tools.
- Automated risk detection capabilities.
- Improved academic evaluation workflow

Workflow Improvement

The system will improve workflow by:

- Centralizing academic communication.
- Reducing fragmented information management.
- Automating performance monitoring.
- Simplifying academic decision support

Benefits of AI Integration

AI integration significantly enhances the platform's intelligence and usability.

Major benefits include:

- Personalized course recommendation.
- Predictive performance analytics.
- Automated student risk detection.
- Real-time campus chatbot assistance.

Instead of functioning as a traditional campus portal, the system evolves into an intelligent academic guidance and student success platform.

Submitted by • CSE4204-8A-T07 • Section 8A

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